

Panel MS-AR(1) with random-effects intercepts, full sample

Pulled real GDP per worker

1 Model

Joint panel Markov-switching AR(1) around a *linear* trend, with three regimes:

$$y_{it} = a_i + g t + z_{it}, \quad (1)$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it}) z_{it} + \sigma(s_{it}) \varepsilon_{it}, \quad (2)$$

$$s_{i,t+1} \sim \Pi(\cdot | s_{it}), \quad (3)$$

$$a_i \sim \mathcal{N}(\alpha, \omega^2) \quad (\text{random effects}). \quad (4)$$

The outcome is log real GDP per worker from the pulled quarterly panel (constant 2015 USD, market FX). The trend is linear in calendar time t (g per year, common across countries). In the pooled specification $a_i \equiv a$. In the random-effects specification the a_i are i.i.d. Normal draws; α and ω are estimated by maximum likelihood, integrating each country's Hamilton-filter likelihood by 11-point Gauss-Hermite quadrature. Cycles are plotted using the posterior mean of a_i . Latent paths s_{it} are country-specific. The regime dated t governs the transition from z_t to z_{t+1} . σ and ρ switch with the regime. The median μ is pinned at 0. $|\rho| < 0.99$. Standard errors (in parentheses) are delta-method from a numerical Hessian on shared parameters. π is the ergodic distribution of Π . $E[z]$ is the long-run mean of the cycle implied by μ , ρ , and Π .

2 Common intercept and linear trend

Pulled panel, full sample: log real GDP per worker. 101 countries, 7450 observations, 1950-Q1–2026-Q2. Fitted countries: 92. Observations: 6883. Log-likelihood: 14717.68. Ergodic mean of the cycle $E[z] = -0.0460$ (0.0576). Common intercept $a = 10.7145$ (0.0847). Common slope $g = -0.0000$ (0.0009).

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-1.2106 (0.1167)	0.0000 —	0.6749 (0.0528)
σ	0.1209 (0.0029)	0.0267 (0.0005)	0.0113 (0.0003)
ρ	0.9900 (0.0001)	0.9900 (0.0000)	0.9900 (0.0000)
Π_0	0.9348 (0.0091)	0.0549 (0.0085)	0.0103 (0.0034)
Π_1	0.0194 (0.0032)	0.9667 (0.0038)	0.0139 (0.0029)
Π_2	0.0168 (0.0032)	0.0095 (0.0031)	0.9737 (0.0035)
π	0.2195 (0.0238)	0.4545 (0.0327)	0.3261 (0.0338)

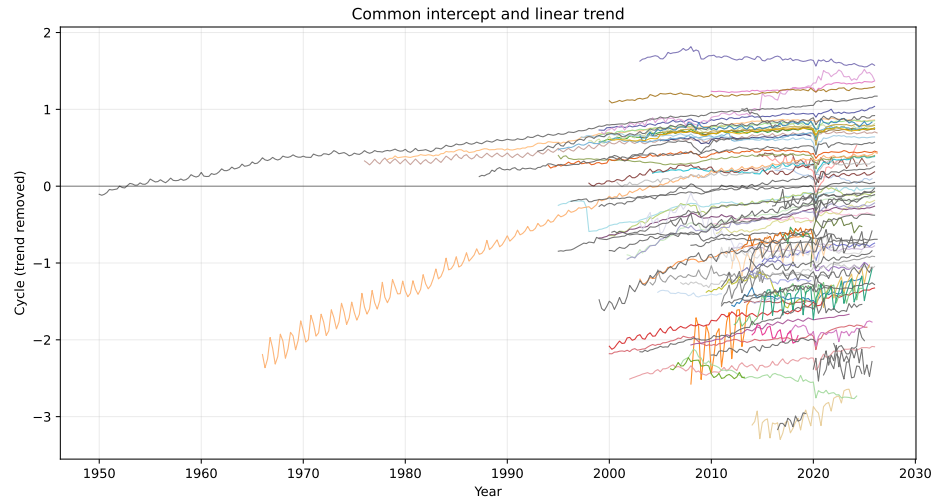


Figure 1: Country cycles after removing the linear trend.

3 Random-effects intercepts, common linear trend

Pulled panel, full sample: log real GDP per worker. 101 countries, 7450 observations, 1950-Q1–2026-Q2. Fitted countries: 92. Observations: 6883. Log-likelihood: 15231.42. Ergodic mean of the cycle $E[z] = -0.2164$ (0.0573). Random intercepts $a_i \sim N(\alpha, \omega^2)$ with $\alpha = 9.5478$ (0.0599), $\omega = 0.6471$ (0.0135). Posterior-mean a_i : mean 9.651, min 7.693, max 11.402. Common slope $g = 0.0105$ (0.0009).

Table 2: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.7694 (0.1496)	0.0000 —	0.0331 (0.0437)
σ	0.1257 (0.0032)	0.0109 (0.0002)	0.0269 (0.0005)
ρ	0.9743 (0.0094)	0.9900 (0.0000)	0.9897 (0.0005)
Π_0	0.9347 (0.0095)	0.0140 (0.0049)	0.0514 (0.0090)
Π_1	0.0127 (0.0028)	0.9731 (0.0034)	0.0142 (0.0034)
Π_2	0.0128 (0.0027)	0.0183 (0.0033)	0.9689 (0.0040)
π	0.1631 (0.0209)	0.3894 (0.0331)	0.4474 (0.0323)

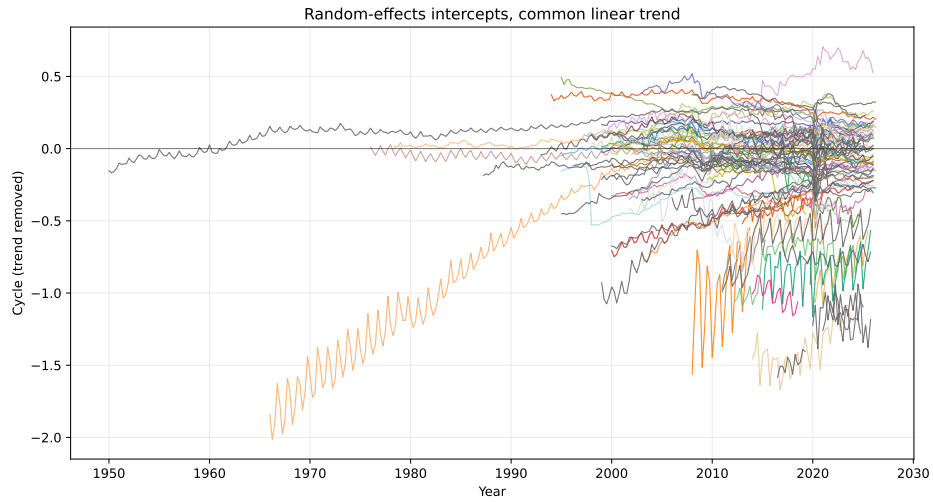


Figure 2: Country cycles after removing the linear trend.